LOGAN THOMAS

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EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES (UCLA) | Bachelor of Science, Mechanical Engineering | GPA: 4.0 — Expected Jun 2029

SKILLS

Design & Modeling: SolidWorks (CSWA), AutoCAD, Design for Manufacturability (DFM), GD&T / Tolerance Analysis, Rapid Prototyping

Simulation & Analysis: FEA (SolidWorks Simulation), MATLAB, Simulink / Simscape (Model Calibration), Python (Data Analysis)

Data & Machine Learning: scikit-learn (Random Forests), Model Validation (Cross-Validation, ROC-AUC), MATLAB Image Analysis

Manufacturing: Manual Machining, Waterjet, Laser Cutting, Sheet Metal Forming, Silicone Molding, Mechanical Assembly

Electronics & Testing: Embedded Systems (Arduino), IMU Integration (MPU-6050), Design of Experiments (DOE), Root Cause Analysis (RCA), Instrumented Thermal Testing, Technical Documentation

EXPERIENCE

ENGINEER INTERN | DOER Marine | *Manned Research Submersible* — Jun 2026 – Present

- **Deployment Rig Design:** Took a moonpool transducer deployment rig from rough concept to a manufacturable SolidWorks assembly, specifying lead screw, bearing, and guide-shaft hardware and resolving mating and isolation constraints.
- **Structural Analysis & Load Path:** Caught a bending-stress concern on the rig's load-bearing frame by hand calculation, then compared three beam configurations in FEA and traced it to an unsupported span, cutting peak stress from 6.57 to 1.27 ksi.
- **Thermal Test Rig & Instrumentation:** Designed and built an instrumented test fixture benchmarking two battery-wall cooling designs across 8 controlled trials, iterating a camera enclosure to eliminate thermal-reflection measurement artifacts.
- **Simulation & Drawings:** Calibrated a Simscape/MATLAB heat-transfer model against trial data (convection 50 → 70 W/m^2K), and produced the full machinist drawing package for a battery enclosure, toleranced for machinability while preserving fit.

ENGINEERING RESEARCH INTERN | I^2BL (UCLA Bioelectronics Lab) | *Ferrobotic Viral Testing Platform* — Jan 2026 – Present

- **Injection Port Integration:** Developed a mass-producible molding process embedding custom 2-mm self-sealing RTV silicone ports into testing cartridges, using root-cause analysis of mold geometry and fill sequencing to prevent recurring air-entrapment defects.
- **Precision Enclosure:** Engineered a modular enclosure with a precision-guided sliding cartridge, achieving sub-millimeter vertical tolerances to preserve optical line-of-sight and unrestricted ferrobot kinematics.

Independent Research Project — Microfluidic Dispenser Characterization & ML Modeling

- **End-to-End ML Pipeline:** Built an end-to-end characterization pipeline spanning experimental design, chip fabrication, 369-trial data collection, and MATLAB image analysis to map dispenser geometry to droplet output.
- **Predictive Modeling & Validation:** Trained a Python random-forest model predicting droplet volume to 0.369 µL cross-validated error, diagnosing overfitting via the CV/train ratio and building an inverse-design tool for geometry selection.

TECHNICAL PROJECTS

SELF-DEPLOYING LAPTOP STAND | Personal Project | *Mechanism Design* — Jun 2026 – Present

- **First-Principles Statics:** Derived the statics of a spring-deployed parallelogram linkage symbolically, proving by contradiction that no equilibrium exists at any deployed angle and reframing the hard stop, not the linkage or spring, as the true load path.
- **FEA Validation & Iteration:** Built the FEA model with pin connectors and compression-only contact at the stop, using mesh convergence to confirm a real stress concentration, then cut platform mass 46% via an FEA-driven rib iteration.
- **Load-Support Architecture:** Redesigned load transfer from linkage-integral stops to a drop-down compression flap after recognizing a single-arc linkage cannot lock against a passive stop without passing dead center, loading PETG in compression rather than bending where it creeps.
- **Materials Testing:** Designed and ran a 12-sample bend test on printed PETG, quantifying ±1.6% measurement uncertainty before claiming significance and finding infill pattern drives stiffness ~3× more than density, then thickened walls instead of raising infill.
- **Print Orientation Validation:** Reasoned out FDM layer-bond anisotropy from first principles, predicted the stiffer print orientation, then confirmed it in a controlled test at −14% deflection while catching a seating artifact that made stiffness numbers method-dependent by up to 60%.